

SIFAT HASAN

Aerospace & Mechanical Engineer | Signal Processing · Controls · Applied ML
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RESEARCH INTERESTS

Active noise & vibration control, acoustic signal processing, adaptive/optimal control, sensor integration and data acquisition systems, applied machine learning for engineering systems.

EDUCATION

Master of Engineering (MEng), Aerospace Engineering

Jan 2022 – May 2023

Toronto Metropolitan University, Toronto, ON

- Best Demo Award, IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (2023)

Bachelor of Engineering (BEng), Mechanical Engineering

Sep 2017 – Jun 2021

China University of Mining and Technology, Xuzhou, China

- 'A' Class Full Tuition Fee Scholarship (2018 – 2021)
- 1st Prize, Structural Design Competition, Jiangsu Province, China (2019)
- Organizing Committee Member & Graphics Designer, Hult Prize at CUMT (2020)

RESEARCH EXPERIENCE

Research Assistant (Full-time)

Apr 2022 – May 2025

Toronto Metropolitan University Aerospace Engineering Centre (RAEC), Downsview, ON

- Designed and developed automated testing methods for aircraft active noise cancellation (ANC) systems, improving data collection efficiency and repeatability
- Built and maintained Python-based testing frameworks for data acquisition, control, and analysis
- Programmed and validated ANC system performance through signal processing and experimental testing
- Configured sensors, data acquisition systems, and integrated hardware/software setups for acoustic experiments
- Conducted data analysis and system validation to ensure accuracy and consistency of results
- Collaborated with research teams and industry partners to troubleshoot and optimize system performance
- Documented system designs, test procedures, and experimental results for technical reporting

PROFESSIONAL EXPERIENCE

Technology Improvement Project Coordinator

Sep 2023 – May 2026

Mississauga Community Legal Services, Mississauga, ON

- Provided technical support across hardware, software, networking, and system access, resolving issues in a fast-paced environment
- Administered Microsoft Dynamics CRM and managed user accounts, permissions, and system configurations
- Led IT infrastructure setup for office relocation, including workstation deployment, network configuration, and systems integration
- Implemented SIP phone system and configured meeting rooms to streamline communication and collaboration
- Developed automation scripts and improved workflows, increasing efficiency and reducing manual effort
- Rebuilt and maintained the organization's website, improving functionality and usability
- Maintained IT asset inventory and created technical documentation for systems and processes

PUBLICATIONS

Adaptive Zone-Based Active Noise Control for a Moving Target. The Journal of the Acoustical Society of America, vol. 155, issue 3_Supplement, pp. A197 [DOI: 10.1121/10.0027287](#)

Aircraft Cabin Active Noise Cancellation Mechanism Using an Actuator Gimbal System. Master's Thesis, Toronto Metropolitan University [DOI: 10.32920/26862367.v1](#)

Integration of Mechanical Systems with IoT. IARJSET: International Advanced Research Journal in Science, Engineering and Technology, vol. 8, issue 1 [DOI: 10.17148/IARJSET.2021.8108](https://doi.org/10.17148/IARJSET.2021.8108)

Additive Manufacturing Using Metal 3D Printers. IARJSET: International Advanced Research Journal in Science, Engineering and Technology, vol. 8, issue 2 [DOI: 10.17148/IARJSET.2021.8217](https://doi.org/10.17148/IARJSET.2021.8217)

SELECTED PROJECTS

Image-to-Image Translation of Satellite Imagery via Conditional GAN

- Trained and evaluated a conditional GAN (Pix2Pix) for satellite-to-feature-map translation, using the UC Berkeley Pix2Pix map dataset (2,194 image pairs), to assess model performance and limitations

TECHNICAL SKILLS

Programming	Python (automation, scripting, data analysis), C++, Arduino (embedded systems)
Engineering & Analysis	SolidWorks, AutoCAD, Ansys, Finite Element Methods (FEM), composite materials QA, data acquisition (DAQ), signal processing, experimental analysis
Systems & IT	System configuration, networking & troubleshooting, CRM/database administration, user access control
Platforms	Microsoft 365, Google Workspace, Adobe Creative Suite (Photoshop, Illustrator, Premiere Pro)

CERTIFICATIONS

- IBM Applied AI Professional Certificate — Coursera
- Control of Mobile Robots — Georgia Tech (Coursera)
- Python Programming & Data Handling — University of Michigan (Coursera)
- Computer Vision with OpenCV — Coursera Project Network
- Google Digital Marketing Certification